How Humans Learn,

The same loop, running on different machines.

What this guide is about

Repeat an action many times and it becomes automatic. That single idea — *pattern formed by repetition + feedback* — explains how a child learns to walk, how a yogi builds a discipline, and how a robot in NVIDIA Isaac Sim learns to balance.

This document explains both sides in plain language, shows where they are the same, and where they are still very different.

Inside this guide

1. The One Big Idea

Strip away all the science vocabulary and learning, in any system, comes down to one loop:

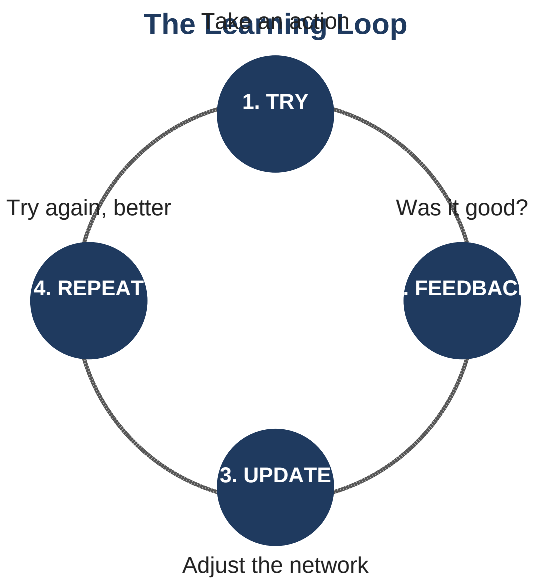


Figure 1. The basic learning loop. Every learning system — biological or artificial — runs some version of this cycle.

A system tries something. The world (or a teacher, or a reward signal) tells it whether that was good or bad. The system adjusts itself slightly. Then it tries again. Do this enough times and the system becomes good at the task — not because it understands the task in any deep sense, but because the **pattern** of "what to do when" has been carved into its network.

The key insight

The substrate changes — neurons in a brain, weights in a neural network — but the loop is structurally the same.

2. How Humans Learn

The two-stage brain

When you first try anything new — driving a car, writing your name, doing a yoga pose — you use your **prefrontal cortex**. This is the front part of your brain, the part responsible for conscious focus and effortful thinking. It's slow, it's tiring, and it can only handle one thing at a time. This is why a new skill feels so exhausting.

But your brain is a lazy genius. It doesn't want to keep paying that high cost. So as you repeat the action, control quietly shifts to a deeper, older structure called the **basal ganglia** — the brain's habit machinery. The basal ganglia don't think. They don't deliberate. They just run patterns. Once a behavior moves here, you can do it while thinking about something else.

In plain words

Prefrontal cortex = the worker who learns the job slowly and consciously.
Basal ganglia = the worker who takes over once the job is memorised, and does it on autopilot.

The role of dopamine: nature's reward signal

Why does the brain bother to lock in some behaviors and forget others? The answer is a chemical called **dopamine**. When something goes well — you taste something delicious, you get a kind word, you nail a difficult move — your brain releases a small pulse of dopamine. That pulse is essentially a signal that says: *"Whatever you just did, do that again."*

The dopamine pulse strengthens the exact neural connections that fired just before the reward. Repeat this enough times and those connections become a thick, well-worn path. That's a habit.

The neuroplasticity principle

Neuroscientists call this **neuroplasticity** — the brain's ability to physically rewire itself based on what it does most often. The famous shorthand is: *neurons that fire together, wire together.*

Three ingredients drive this rewiring:

- **Repetition** — the same neural path firing over and over.
- **Reward** — a dopamine signal saying "yes, more of that."
- **Reflection** — pausing to think about what worked, which engages the prefrontal cortex and helps consolidate the learning.

This is why a skilled musician can play a piece without thinking, why driving home from work feels like it happens by itself, and why bad habits are so hard to break — the wiring is literally there in the brain.

3. How Robots Learn

The old way vs the new way

Traditional robots ran on **hand-written rules**. An engineer would type out instructions like "if the obstacle is closer than 0.5 metres, stop." This works for predictable factories but breaks down the moment the world changes — a new object shape, a different surface, unusual lighting.

Modern robots learn instead. They are given a goal and allowed to figure out their own behavior through trial and error. The result is not a list of rules a human can read — it is a **neural network policy**: a giant mathematical pattern that maps "what I'm seeing right now" to "what I should do next."

Reinforcement learning in plain words

The training method behind most modern robot brains is called **reinforcement learning (RL)**. The recipe is remarkably simple:

1. The robot tries an action in its environment.
2. The environment returns a **reward** — a number saying how good that action was.
3. The robot's neural network adjusts its weights very slightly, making good actions a tiny bit more likely next time.
4. Repeat — millions of times.

The mapping back to humans

The robot's "reward" plays the role of dopamine. Its "weight adjustment" plays the role of neural rewiring. Its "trying millions of times" plays the role of practice. Different words. Same loop.

Isaac Sim: the practice arena

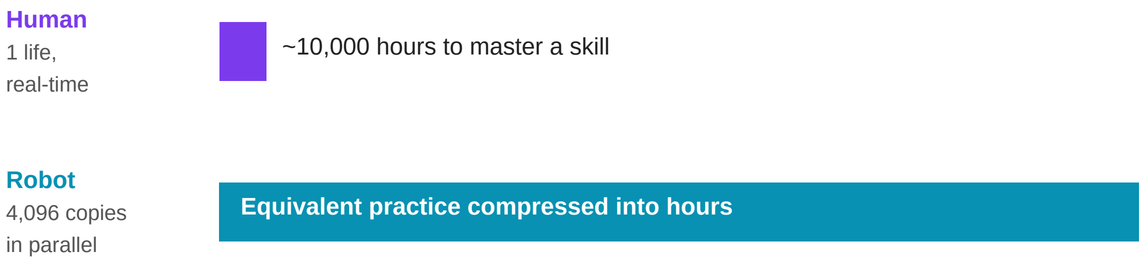
Letting a real robot fail millions of times is impossibly expensive — and dangerous. It would break itself, break things around it, and take years. So the modern approach is to train inside a **simulator** first: a virtual physics world where a digital copy of the robot can practice safely.

NVIDIA Isaac Sim is the most popular of these. It is a high-quality virtual world with realistic physics, lighting, and sensor noise. **Isaac Lab** is the training framework that sits on top of it and runs the actual reinforcement learning.

The clever trick: Isaac Lab doesn't run just one robot. It runs **thousands of copies of the same robot in parallel**, all on a single graphics card, all practicing the same task at the same time.

Every copy contributes its experience to one shared brain. What would take a single robot months of practice can be compressed into a few hours.

Speed: Why Robots Can Out-Practice Humans



Same learning principle. Different speed.

Figure 2. A simulator's main superpower is parallelism — many copies, one shared lesson.

From simulation to the real world ("sim-to-real")

A robot that masters a task in simulation may still fail in the real world, because simulators are never perfectly accurate. The fix is a technique called **domain randomisation**: deliberately mess with the simulation. Random friction, random lighting, random object weights, random sensor glitches. The robot is forced to learn patterns robust enough to survive surprises — and that robustness carries over to real hardware.

This is analogous to a human who practiced batting only on a perfect indoor pitch, versus one who practiced in wind, in rain, with different bats, on different surfaces. The second player will adapt much better on game day.

4. Side-by-Side: Where Humans and Robots Overlap

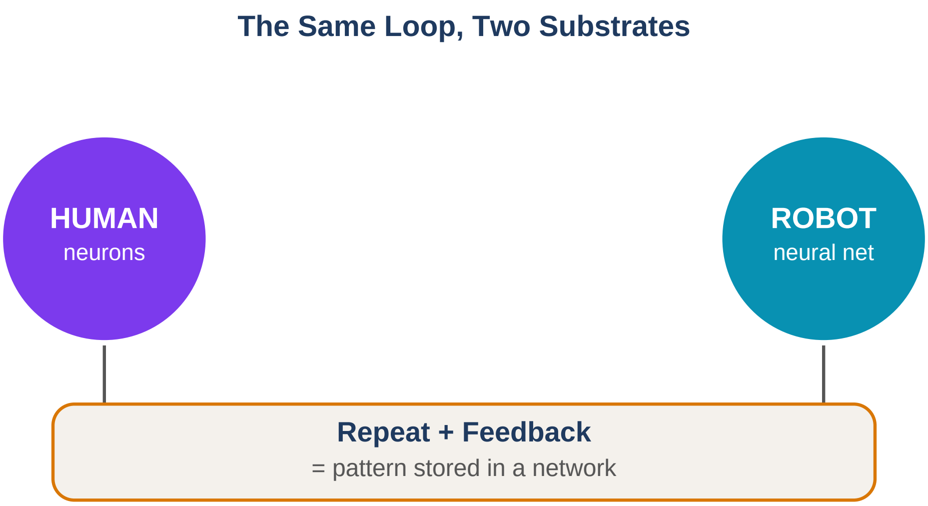


Figure 3. Two very different machines running the same underlying loop.

Once you see the loop, the parallels become hard to unsee. Here is the same idea translated across both worlds:

Step in the loop	Human	Robot in Isaac Sim
Try an action	Move your body, attempt a task	Output motor commands to the simulated robot
Get feedback	Outcome — pain, pleasure, success, failure	Reward signal — a number from the reward function
Chemical / mathematical signal	Dopamine pulse from the brain's reward system	Gradient computed by the RL algorithm
Update the network	Neural connections strengthen or weaken	Neural network weights adjust slightly
Repeat	Hours, days, years of practice	Millions of episodes, often in parallel
End result	A habit, a skill, a reflex	A trained policy file

The deepest similarity is conceptual: in both cases, **nobody writes the final behavior down**. There is no rule book inside your basal ganglia, and there is no if-then list inside a trained robot policy. The skill exists as a distributed pattern across a vast network of connections.

5. Re-Learning: Changing a Pattern That's Already Burned In

Both systems face the same problem when they need to change. The old pattern is already stored in the network. You cannot simply delete it. The only way out is to **train a competing pattern strongly enough that it wins the race** when the trigger fires.

The challenge	How humans handle it	How robots handle it
Notice the old pattern	Mindful awareness; catching yourself mid-habit	Diagnose which part of the policy is failing
Interrupt it	Conscious effort, often uncomfortable	Stop using the current policy
Build the new path	Deliberate repetition of the new behavior	Fine-tune the network on new data
Avoid forgetting old skills	Don't abandon useful prior habits	Replay buffers; gentle fine-tuning
Use environment	Change context — new place, new triggers	Curriculum learning — gradually shift the simulation
Time required	Weeks to months of consistent practice	Hours to days of additional training

Researchers call the robot version of this problem **catastrophic forgetting** — if you train too aggressively on a new skill, the old ones get overwritten. In humans, the same thing happens more gently: old habits leave traces and tend to resurface under stress, fatigue, or familiar surroundings. This is why recovering addicts relapse in old neighborhoods, and why you fall into old speech patterns around family.

The shared rule

You cannot delete a pattern. You can only outweigh it with consistent reps of a better one. This is true for the basal ganglia and for a neural network policy alike.

6. Where the Analogy Breaks

It would be easy to conclude that humans and robots are the same kind of thing — just running on different hardware. The mechanics are similar enough to make that tempting. But there are at least three places where the comparison becomes thinner.

1. Robots have no inner motivation to change

A robot does not wake up one day and decide it would like to learn a new skill. An engineer decides that. The robot has no preference about which patterns it carries. A human, sometimes, can generate the intention to change a pattern from inside — and that intention itself is part of how the change happens.

2. Humans can watch their own patterns running

You can be in the middle of an angry reaction and notice that you are being angry. That noticing slightly changes the pattern. A current robot policy has no such self-observation — it pattern-matches input to output without any inner witness. Whether this self-watching capacity is itself just another pattern, or something categorically different, is still an open question in both neuroscience and philosophy.

3. There is something it is like to be a human

A thermostat runs a pattern. A trained robot runs a pattern. Neither, as far as we can tell, *feels* like anything from the inside. Whatever you are doing right now reading this — there is a subjective experience of doing it. That felt quality is not explained by the loop diagram. It is what philosophers call the "hard problem of consciousness," and no one has solved it.

A useful framing from Indian thought

Yoga and Advaita traditions describe a useful distinction: *prakriti* (the patterned, conditioned, learning machinery) and the awareness in which those patterns appear. From this view, the answer is not "you are the patterns" or "you are something outside them" — it is that the patterns are real, the awareness is real, and the mistake is confusing the awareness for the patterns. Whatever does the watching is the question worth sitting with.

7. Takeaways

Learning is a loop. Try → feedback → adjust → repeat. Every learning system runs some version of this.

Repetition writes the pattern. There is no shortcut. Neither neurons nor neural networks update without reps.

Reward decides what is kept. Dopamine in humans, reward functions in robots. They serve the same purpose: tagging what to do more of.

The skill is not written down. In both cases, the behavior lives as a distributed pattern across the network, not as a readable rule.

Re-learning is hard, not impossible. Old patterns persist. The only way to change them is to build a stronger competing pattern, with enough reps and the right environment.

Simulation is the modern training ground. What used to require real-world trial and error can now happen in virtual environments like Isaac Sim — faster, safer, and at massive parallel scale.

Mechanism is not the whole story. Humans appear to have something robots currently lack: the ability to notice patterns running and choose differently. Whether this is just another pattern or something more is genuinely unknown.

One sentence to remember

Whether neurons or weights, the rule is the same — what you practice, the network becomes.

Sources consulted
Yin & Knowlton, "The role of the basal ganglia in habit formation," *Nature Reviews Neuroscience* · Western University Thrive — The Science Behind Habits · Harvard Health — Brain plasticity in drug addiction · NYU Langone — Dopamine and acetylcholine in learning · NVIDIA Isaac Lab documentation — Getting Started with Robot Learning · NVIDIA Use Case: Robot Learning in Simulation · Salimpour et al., "Sim-to-Real Transfer for Mobile Robots with Reinforcement Learning," arXiv:2501.02902.